that the planets revolved around the Sun—but no explanation existed for *why* the planets moved in orbit. Newton solved this problem. From his observations in the orchard, he concluded that there must be an invisible force pulling the apple (and therefore all objects with mass) to the Earth. He further theorized that the force affecting movement of the apple must also affect celestial bodies, such as the Sun, Moon, and planets. Newton became the first scientist to understand that it was gravity that controlled the movement of the planets and held them in orbit.

Newtonian Laws

In 1687, Newton published his findings in his landmark text *Philosophiae Naturalis Principia Mathematica*, one of the most important books in scientific history. In it, he describes his three laws of motion, which act as a mathematical explanation of how speed and mass affect the movement of an object. The laws state that a static object will move if force is exerted on it; that acceleration depends on the force exerted and the mass of the object; and that all forces act in pairs that are equal and opposite. His law of gravity states that gravity is a universal force and that all objects (including celestial ones) exert a gravitational pull on each other. This represented one of the most significant scientific breakthroughs of all time.

Other fields

Newton extensively studied the nature of light and made great advances, which he published in his 1704 text *Optiks*. Through experiments with clear glass prisms, he revealed that white light is composed of all colors of the spectrum. He also pioneered the use of light through telescopes by designing the first reflecting telescope in 1668, which used mirrors (instead of lenses) to reflect light and create a sharper image than had previously been possible.

Beyond science, Newton applied himself to the study of alchemy and reinterpretations of the Bible.

CHRISTIAAN **HUYGENS**

An influential figure in the 17th-century scientific revolution, Christiaan Huygens was a Dutch scientist and mathematician whose discoveries in physics and astronomy left a lasting legacy.

After studying law and mathematics, Huygens (1629–1695) became interested in astronomy and optics. He built telescopes to improve visual accuracy and discovered Saturn's largest moon, as well as correctly describing its rings. He founded the theory that light travels like a wave and explored forces and motion; his work coincided with Newton's, yet they disagreed about light and gravity. Huygens also discovered the pendulum as a regulator of clock movement.